

NVIDIA Corp.

All Signs Still Point to Another Revision Higher

Our checks continue to point to billion+ of upside in April/July with calls for an air pocket seemingly overblown.

Asia checks point to NVDA DC GPU unit upside potential close to 10% in the April quarter (vs. run rate) and just over 20% in the July quarter as capacity continues to come online. In recent checks, the supply chain continues to expect about the same unit upside for April and July, but we believe the company can capture incremental revenue upside due to pricing on H200, which will start shipping in the July quarter. B-series hand off still expected later in the year, likely Q4. Looking into next year, the supply chain continues to expect GB systems to launch ahead of standalone Blackwell GPUs and expects H-series to continue to be a healthy portion of overall mix into next year, in-line with management commentary from our facility visit last quarter (note [here](#)). While investor concerns through the quarter have largely centered on a potential air pocket in the middle of the year ahead of the Blackwell launch, we still do not see evidence of this occurring, with the tone from our checks remaining mostly unchanged. All in, checks continue to point to >\$1B in upside in April and \$2B in July. We see DC ahead in April and now believe the segment ultimately comes in closer to \$23B (with ~\$3.5B coming from MLNX) vs. St. of \$21.1B. For July, we see additional DC upside and expect there to be enough upside for DC revenue of \$24.5B (vs. St. \$22.8B, with a MLNX contribution of ~\$3.75B). Looking forward, the next major step up in revenue will be from the ramp of the GB200. The company intra-Q spoke very positively of the total mix moving towards the complete GB200 solution for hyperscalers, and we don't believe this mix shift is fully appreciated.

NVDA: Quarterly and Annual EPS (USD)

	2024	2025			2026			Change y/y	
FY Jan	Actual	Old	New	Cons	Old	New	Cons	2025	2026
Q1	1.09A	5.57E	5.57E	5.54E	6.68E	6.68E	7.20E	411%	20%
Q2	2.70A	5.81E	5.81E	6.00E	6.84E	6.84E	7.55E	115%	18%
Q3	4.02A	6.12E	6.12E	6.54E	6.97E	6.97E	7.89E	52%	14%
Q4	5.16A	6.43E	6.43E	7.11E	7.10E	7.10E	8.31E	25%	10%
Year	12.96A	23.93E	23.93E	25.24E	27.59E	27.59E	31.21E	85%	15%
P/E	71.4		38.6			33.5			

Consensus numbers are from Bloomberg received on 17-May-2024; 12:54 GMT
Source: Barclays Research

Barclays Capital Inc. and/or one of its affiliates does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Please see analyst certifications and important disclosures beginning on page 3.

Completed: 19-May-24, 23:58 GMT Released: 20-May-24, 04:10 GMT Restricted - External

CORE

NVDA

OVERWEIGHT

Unchanged

U.S. Semiconductors & Semiconductor Capital Equipment

NEUTRAL

Unchanged

Price Target

USD 1100.00

raised 29% from USD 850.00

Price (17-May-24)

USD 924.79

Potential Upside/Downside

+18.9%

Market Cap (USD mn)

2274830

Shares Outstanding (mn)

2459.83

Free Float (%)

96.00

52 Wk Avg Daily Volume (mn)

48.8

Dividend Yield (%)

0.02

Return on Equity TTM (%)

91.46

Current BVPS (USD)

17.44

Source: Bloomberg

Price Performance

Exchange-Nasdaq

52 Week range

USD 974.00-294.30



Source: IDC
[Link to Barclays Live for interactive charting](#)

U.S. Semiconductors & Semiconductor Capital Equipment

Tom O'Malley

+1 212 526 0692

thomas.o'malley@barclays.com

BCI, US

Scott Fessler

+1 212 526 2604

scott.fessler@barclays.com

BCI, US

William Levy

+1 212 526 8385

william.levy@barclays.com

BCI, US

U.S. Semiconductors & Semiconductor Capital Equipment

NEUTRAL

NVIDIA Corp. (NVDA)

OVERWEIGHT

Income statement (\$mn)	2024A	2025E	2026E	2027E	CAGR
Revenue	60,922	105,479	121,584	N/A	N/A
EBITDA (adj)	38,642	70,899	80,826	N/A	N/A
EBIT (adj)	37,134	69,351	79,278	N/A	N/A
Pre-tax income (adj)	37,745	70,501	80,828	N/A	N/A
Net income (adj)	32,312	59,591	68,704	N/A	N/A
EPS (adj) (\$)	12.96	23.93	27.59	N/A	N/A
Diluted shares (mn)	2,493	2,490	2,490	N/A	N/A
DPS (\$)	0.16	0.16	0.16	N/A	N/A
Margin and return data	2024A	2025E	2026E	2027E	Average
EBITDA (adj) margin (%)	63.4	67.2	66.5	N/A	65.7
EBIT (adj) margin (%)	61.0	65.7	65.2	N/A	64.0
Pre-tax (adj) margin (%)	62.0	66.8	66.5	N/A	65.1
Net (adj) margin (%)	53.0	56.5	56.5	N/A	55.3
ROA (%)	78.5	90.7	63.2	N/A	77.4
ROE (%)	146.2	138.7	80.9	N/A	121.9
Balance sheet and cash flow (\$mn)	2024A	2025E	2026E	2027E	CAGR
Tangible fixed assets	3,914	3,966	4,218	N/A	N/A
Intangible fixed assets	4,430	4,430	4,430	N/A	N/A
Cash and equivalents	25,984	62,979	112,563	N/A	N/A
Total assets	65,728	108,789	160,245	N/A	N/A
Short and long-term debt	9,709	9,709	9,709	N/A	N/A
Other long-term liabilities	2,541	2,541	2,541	N/A	N/A
Total liabilities	22,750	23,822	24,191	N/A	N/A
Net debt/(funds)	-16,275	-53,270	-102,854	N/A	N/A
Shareholders' equity	42,978	84,967	136,054	N/A	N/A
Change in working capital	17,204	42,669	N/A	N/A	N/A
Cash flow from operations	28,091	56,358	69,146	N/A	N/A
Capital expenditure	-1,068	-1,600	-1,800	N/A	N/A
Free cash flow	27,023	54,758	67,346	N/A	N/A
Valuation and leverage metrics	2024A	2025E	2026E	2027E	Average
P/E (adj) (x)	71.4	38.6	33.5	N/A	47.8
EV/sales (x)	37.5	21.3	18.1	N/A	25.7
EV/EBITDA (adj) (x)	59.2	31.7	27.2	N/A	39.4
Equity FCF yield (%)	1.2	2.4	2.9	N/A	2.2
P/Sales (x)	37.8	21.8	18.9	N/A	26.2
P/BV (x)	53.6	27.1	16.9	N/A	32.6
Dividend yield (%)	0.0	0.0	0.0	N/A	0.0

Note: FY End Jan

Source: Company data, Bloomberg, Barclays Research

Price (17-May-2024) **USD 924.79**
 Price Target **USD 1,100.00**

Why OVERWEIGHT?

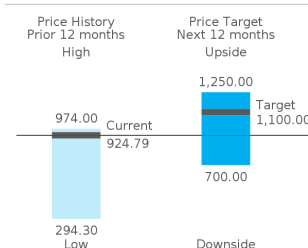
We are OW as the company has long-term sustainable growth led by a large lead in GPUs for AI in DC, with further Edge opportunities (Autos, robots, etc.) and a competitive moat around a large portion of the market.

Upside case **USD 1,250.00**

Our upside case of \$1250 is based upon 42x our upside CY25E non-GAAP EPS of \$30.00, which assumes larger TAM for AI in the DC, higher Networking attach, and higher GMs.

Downside case **USD 700.00**

Our downside case of \$700 is based upon 30x our downside CY25E non-GAAP EPS of \$23.00, which assumes a correction of AI spending in the DC, a slower Auto ramp and pricing/share pressure.

Upside/Downside scenarios

Changes to PT

Our price target of \$1100 is based upon 40x our CY25E non-GAAP EPS of \$27.59. Our prior price target of \$850 was based on 31x our CY25E non-GAAP EPS of \$27.59

Analyst(s) Certification(s):

I, Tom O'Malley, hereby certify (1) that the views expressed in this research report accurately reflect my personal views about any or all of the subject securities or issuers referred to in this research report and (2) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report.

Important Disclosures:

Barclays Research is produced by the Investment Bank of Barclays Bank PLC and its affiliates (collectively and each individually, "Barclays"). All authors contributing to this research report are Research Analysts unless otherwise indicated. The publication date at the top of the report reflects the local time where the report was produced and may differ from the release date provided in GMT.

Availability of Disclosures:

Where any companies are the subject of this research report, for current important disclosures regarding those companies please refer to <https://publicresearch.barclays.com> or alternatively send a written request to: Barclays Research Compliance, 745 Seventh Avenue, 13th Floor, New York, NY 10019 or call +1-212-526-1072.

The analysts responsible for preparing this research report have received compensation based upon various factors including the firm's total revenues, a portion of which is generated by investment banking activities, the profitability and revenues of the Markets business and the potential interest of the firm's investing clients in research with respect to the asset class covered by the analyst.

Analysts regularly conduct site visits to view the material operations of covered companies, but Barclays policy prohibits them from accepting payment or reimbursement by any covered company of their travel expenses for such visits.

Barclays Research Department produces various types of research including, but not limited to, fundamental analysis, equity-linked analysis, quantitative analysis, and trade ideas. Recommendations contained in one type of Barclays Research may differ from those contained in other types of Barclays Research, whether as a result of differing time horizons, methodologies, or otherwise.

In order to access Barclays Statement regarding Research Dissemination Policies and Procedures, please refer to <https://publicresearch.barcap.com/S/RD.htm>. In order to access Barclays Research Conflict Management Policy Statement, please refer to: <https://publicresearch.barcap.com/S/CM.htm>.

Primary Stocks (Ticker, Date, Price)

NVIDIA Corp. (NVDA, 17-May-2024, USD 924.79), Overweight/Neutral, CD/CE/J/K/M

Unless otherwise indicated, prices are sourced from Bloomberg and reflect the closing price in the relevant trading market, which may not be the last available closing price at the time of publication.

Disclosure Legend:

A: Barclays Bank PLC and/or an affiliate has been lead manager or co-lead manager of a publicly disclosed offer of securities of the issuer in the previous 12 months.

B: An employee or non-executive director of Barclays PLC is a director of this issuer.

CD: Barclays Bank PLC and/or an affiliate is a market-maker in debt securities issued by this issuer.

CE: Barclays Bank PLC and/or an affiliate is a market-maker in equity securities issued by this issuer.

CH: Barclays Bank PLC and/or its group companies makes, or will make, a market in the securities (as defined under paragraph 16.2 (k) of the HK SFC Code of Conduct) in respect of this issuer.

D: Barclays Bank PLC and/or an affiliate has received compensation for investment banking services from this issuer in the past 12 months.

E: Barclays Bank PLC and/or an affiliate expects to receive or intends to seek compensation for investment banking services from this issuer within the next 3 months.

FA: Barclays Bank PLC and/or an affiliate beneficially owns 1% or more of a class of equity securities of this issuer, as calculated in accordance with US regulations.

FB: Barclays Bank PLC and/or an affiliate beneficially owns a long position of more than 0.5% of a class of equity securities of this issuer, as calculated in accordance with EU regulations.

FC: Barclays Bank PLC and/or an affiliate beneficially owns a short position of more than 0.5% of a class of equity securities of this issuer, as calculated in accordance with EU regulations.

FD: Barclays Bank PLC and/or an affiliate beneficially owns 1% or more of a class of equity securities of this issuer, as calculated in accordance with South Korean regulations.

FE: Barclays Bank PLC and/or its group companies has financial interests in relation to this issuer and such interests aggregate to an amount equal to or more than 1% of this issuer's market capitalization, as calculated in accordance with HK regulations.

GD: One of the Research Analysts on the fundamental credit coverage team (and/or a member of his or her household) has a long position in the common equity securities of this issuer.

GE: One of the Research Analysts on the fundamental equity coverage team (and/or a member of his or her household) has a long position in the common equity securities of this issuer.

H: This issuer beneficially owns more than 5% of any class of common equity securities of Barclays PLC.

I: Barclays Bank PLC and/or an affiliate is party to an agreement with this issuer for the provision of financial services to Barclays Bank PLC and/or an affiliate.

J: Barclays Bank PLC and/or an affiliate is a liquidity provider and/or trades regularly in the securities of this issuer and/or in any related derivatives.

K: Barclays Bank PLC and/or an affiliate has received non-investment banking related compensation (including compensation for brokerage services, if applicable) from this issuer within the past 12 months.

L: This issuer is, or during the past 12 months has been, an investment banking client of Barclays Bank PLC and/or an affiliate.

M: This issuer is, or during the past 12 months has been, a non-investment banking client (securities related services) of Barclays Bank PLC and/or an affiliate.

N: This issuer is, or during the past 12 months has been, a non-investment banking client (non-securities related services) of Barclays Bank PLC and/or an affiliate.

O: Not in use.

P: A partner, director or officer of Barclays Capital Canada Inc. has, during the preceding 12 months, provided services to the subject company for remuneration, other than normal course investment advisory or trade execution services.

Q: Barclays Bank PLC and/or an affiliate is a Corporate Broker to this issuer.

R: Barclays Capital Canada Inc. and/or an affiliate has received compensation for investment banking services from this issuer in the past 12 months.

S: This issuer is a Corporate Broker to Barclays PLC.

T: Barclays Bank PLC and/or an affiliate is providing investor engagement services to this issuer.

U: The equity securities of this Canadian issuer include subordinate voting restricted shares.

V: The equity securities of this Canadian issuer include non-voting restricted shares.

Risk Disclosure(s)

Master limited partnerships (MLPs) are pass-through entities structured as publicly listed partnerships. For tax purposes, distributions to MLP unit holders may be treated as a return of principal. Investors should consult their own tax advisors before investing in MLP units.

Disclosure(s) regarding Information Sources

Copyright © (2024) Sustainalytics. Sustainalytics retains ownership and all intellectual property rights in its proprietary information and data that may be included in this report. Any Sustainalytics' information and data included herein may not be copied or redistributed, is intended for informational purposes only, does not constitute investment advice and is not warranted to be complete, timely and accurate. Sustainalytics' information and data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers/>

Bloomberg® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg") and the Bloomberg Indices are trademarks of Bloomberg. Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Bloomberg does not approve or endorse this material, or guarantee the accuracy or completeness of any information herein, or make any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, Bloomberg shall have no liability or responsibility for injury or damages arising in connection therewith.

Guide to the Barclays Fundamental Equity Research Rating System:

Our coverage analysts use a relative rating system in which they rate stocks as Overweight, Equal Weight or Underweight (see definitions below) relative to other companies covered by the analyst or a team of analysts that are deemed to be in the same industry (the "industry coverage universe").

In addition to the stock rating, we provide industry views which rate the outlook for the industry coverage universe as Positive, Neutral or Negative (see definitions below). A rating system using terms such as buy, hold and sell is not the equivalent of our rating system. Investors should carefully read the entire research report including the definitions of all ratings and not infer its contents from ratings alone.

Stock Rating

Overweight - The stock is expected to outperform the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Equal Weight - The stock is expected to perform in line with the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Underweight - The stock is expected to underperform the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Rating Suspended - The rating and target price have been suspended temporarily due to market events that made coverage impracticable or to comply with applicable regulations and/or firm policies in certain circumstances including where the Investment Bank of Barclays Bank PLC is acting in an advisory capacity in a merger or strategic transaction involving the company.

Industry View

Positive - industry coverage universe fundamentals/valuations are improving.

Neutral - industry coverage universe fundamentals/valuations are steady, neither improving nor deteriorating.

Negative - industry coverage universe fundamentals/valuations are deteriorating.

Below is the list of companies that constitute the "industry coverage universe":

U.S. Semiconductors & Semiconductor Capital Equipment

Advanced Micro Devices (AMD)	Allegro Microsystems (ALGM)	Analog Devices (ADI)
Applied Materials Inc. (AMAT)	Arm Holdings plc (ARM)	Astera Labs, Inc. (ALAB)
Broadcom Inc. (AVGO)	Camtek (CAMT)	Cirrus Logic Inc. (CRUS)
Coherent Corp. (COHR)	Credo Technology Group (CRDO)	Intel Corp. (INTC)
KLA Corporation (KLAC)	Lam Research Corporation (LRCX)	Lumentum Holdings Inc. (LITE)

MACOM Technology Solutions Holdings, Inc. (MTSI)	Marvell Technology Group, Ltd. (MRVL)	Micron Technology, Inc. (MU)
NVIDIA Corp. (NVDA)	NXP Semiconductors NV (NXPI)	Qorvo Inc. (QRVO)
QUALCOMM, Inc. (QCOM)	Seagate Technology plc (STX)	SGH (SGH)
Silicon Laboratories, Inc. (SLAB)	SiTime Corporation (SITM)	Skyworks Solutions, Inc. (SWKS)
Texas Instruments, Inc. (TXN)	Veeco Instruments Inc (VECO)	Western Digital Corporation (WDC)

Distribution of Ratings:

Barclays Equity Research has 1831 companies under coverage.

49% have been assigned an Overweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Buy rating; 50% of companies with this rating are investment banking clients of the Firm; 71% of the issuers with this rating have received financial services from the Firm.

36% have been assigned an Equal Weight rating which, for purposes of mandatory regulatory disclosures, is classified as a Hold rating; 42% of companies with this rating are investment banking clients of the Firm; 62% of the issuers with this rating have received financial services from the Firm.

14% have been assigned an Underweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Sell rating; 24% of companies with this rating are investment banking clients of the Firm; 52% of the issuers with this rating have received financial services from the Firm.

Guide to the Barclays Research Price Target:

Each analyst has a single price target on the stocks that they cover. The price target represents that analyst's expectation of where the stock will trade in the next 12 months. Upside/downside scenarios, where provided, represent potential upside/potential downside to each analyst's price target over the same 12-month period.

Types of investment recommendations produced by Barclays Equity Research:

In addition to any ratings assigned under Barclays' formal rating systems, this publication may contain investment recommendations in the form of trade ideas, thematic screens, scorecards or portfolio recommendations that have been produced by analysts within Equity Research. Any such investment recommendations shall remain open until they are subsequently amended, rebalanced or closed in a future research report.

Barclays may also re-distribute equity research reports produced by third-party research providers that contain recommendations that differ from and/or conflict with those published by Barclays' Equity Research Department.

Disclosure of other investment recommendations produced by Barclays Equity Research:

Barclays Equity Research may have published other investment recommendations in respect of the same securities/instruments recommended in this research report during the preceding 12 months. To view all investment recommendations published by Barclays Equity Research in the preceding 12 months please refer to <https://live.barcap.com/go/research/Recommendations>.

Legal entities involved in producing Barclays Research:

Barclays Bank PLC (Barclays, UK)

Barclays Capital Inc. (BCI, US)

Barclays Bank Ireland PLC, Frankfurt Branch (BBI, Frankfurt)

Barclays Bank Ireland PLC, Paris Branch (BBI, Paris)

Barclays Bank Ireland PLC, Milan Branch (BBI, Milan)

Barclays Securities Japan Limited (BSJL, Japan)

Barclays Bank PLC, Hong Kong Branch (Barclays Bank, Hong Kong)

Barclays Bank Mexico, S.A. (BBMX, Mexico)

Barclays Capital Casa de Bolsa, S.A. de C.V. (BCCB, Mexico)

Barclays Securities (India) Private Limited (BSIPL, India)

Barclays Bank PLC, Singapore Branch (Barclays Bank, Singapore)

Barclays Bank PLC, DIFC Branch (Barclays Bank, DIFC)

NVIDIA Corp. (NVDA / NVDA)

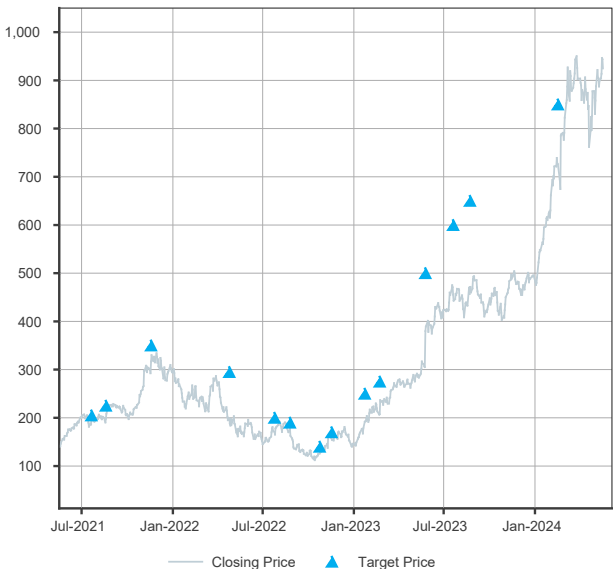
Stock Rating: **OVERWEIGHT**

Industry View: **NEUTRAL**

Closing Price: **USD 924.79** (17-May-2024)

Rating and Price Target Chart - USD (as of 17-May-2024)

Currency=USD



Source: IDC, Barclays Research

[Link to Barclays Live for interactive charting](#)

Publication Date	Closing Price*	Rating	Adjusted Price Target
16-Feb-2024	726.58		850.00
23-Aug-2023	471.16		650.00
20-Jul-2023	470.77		600.00
25-May-2023	305.38		500.00
22-Feb-2023	207.54		275.00
23-Jan-2023	178.39		250.00
17-Nov-2022	159.10		170.00
24-Oct-2022	124.66		140.00
25-Aug-2022	172.22		190.00
25-Jul-2022	173.19		200.00
25-Apr-2022	195.15		295.00
18-Nov-2021	292.61		350.00
19-Aug-2021	197.98		225.00
21-Jul-2021	194.10		205.00

On 19-May-2021, prior to any intra-day change that may have been published, the rating for this security was Overweight, and the adjusted price target was 162.50.

Source: Bloomberg, Barclays Research

*This is the closing price referenced in the publication, which may not be the last available closing price at the time of publication.

Historical stock prices and price targets may have been adjusted for stock splits and dividends.

CD: Barclays Bank PLC and/or an affiliate is a market-maker in debt securities issued by NVIDIA Corp..

CE: Barclays Bank PLC and/or an affiliate is a market-maker in equity securities issued by NVIDIA Corp..

J: Barclays Bank PLC and/or an affiliate is a liquidity provider and/or trades regularly in the securities by NVIDIA Corp. and/or in any related derivatives.

K: Barclays Bank PLC and/or an affiliate has received non-investment banking related compensation (including compensation for brokerage services, if applicable) from NVIDIA Corp. within the past 12 months.

M: NVIDIA Corp. is, or during the past 12 months has been, a non-investment banking client (securities related services) of Barclays Bank PLC and/or an affiliate.

Valuation Methodology: Our price target of \$1100 is based upon 40x our CY25E non-GAAP EPS of \$27.59.

Risks which May Impede the Achievement of the Barclays Research Valuation and Price Target: Risks to the downside include a slower ramp in GPUs for AI in the DC, slower recovery in the gaming market, a slower Auto ramp and slower GM improvement.

Disclaimer:

This publication has been produced by Barclays Research Department in the Investment Bank of Barclays Bank PLC and/or one or more of its affiliates (collectively and each individually, "Barclays").

It has been prepared for institutional investors and not for retail investors. It has been distributed by one or more Barclays affiliated legal entities listed below or by an independent and non-affiliated third-party entity (as may be communicated to you by such third-party entity in its communications with you). It is provided for information purposes only, and Barclays makes no express or implied warranties, and expressly disclaims all warranties of merchantability or fitness for a particular purpose or use with respect to any data included in this publication. To the extent that this publication states on the front page that it is intended for institutional investors and is not subject to all of the independence and disclosure standards applicable to debt research reports prepared for retail investors under U.S. FINRA Rule 2242, it is an "institutional debt research report" and distribution to retail investors is strictly prohibited. Barclays also distributes such institutional debt research reports to various issuers, media, regulatory and academic organisations for their own internal informational news gathering, regulatory or academic purposes and not for the purpose of making investment decisions regarding any debt securities. Media organisations are prohibited from re-publishing any opinion or recommendation concerning a debt issuer or debt security contained in any Barclays institutional debt research report. Any such recipients that do not want to continue receiving Barclays institutional debt research reports should contact debtresearch@barclays.com. Unless clients have agreed to receive "institutional debt research reports" as required by US FINRA Rule 2242, they will not receive any such reports that may be co-authored by non-debt research analysts. Eligible clients may get access to such cross asset reports by contacting debtresearch@barclays.com. Barclays will not treat unauthorized recipients of this report as its clients and accepts no liability for use by them of the contents which may not be suitable for their personal use. Prices shown are indicative and Barclays is not offering to buy or sell or soliciting offers to buy or sell any financial instrument.

Without limiting any of the foregoing and to the extent permitted by law, in no event shall Barclays, nor any affiliate, nor any of their respective officers, directors, partners, or employees have any liability for (a) any special, punitive, indirect, or consequential damages; or (b) any lost profits, lost revenue, loss of anticipated savings or loss of opportunity or other financial loss, even if notified of the possibility of such damages, arising from any use of this publication or its contents.

Other than disclosures relating to Barclays, the information contained in this publication has been obtained from sources that Barclays Research believes to be reliable, but Barclays does not represent or warrant that it is accurate or complete. Appearances by Third-Party Speakers: Any views or opinions expressed by third-party speakers during this event are solely those of the speaker and do not represent the views or opinions of Barclays. Barclays is not responsible for, and makes no warranties whatsoever as to, the information or opinions contained in any written, electronic, audio or video presentations by any third-party speakers at the event ("Third-Party Content"). Any such Third-Party Content has not been adopted or endorsed by Barclays and does not represent the views or opinions of Barclays. Third-Party Content is provided for information purposes only and has not been independently verified by Barclays for its accuracy or completeness.

The views in this publication are solely and exclusively those of the authoring analyst(s) and are subject to change, and Barclays Research has no obligation to update its opinions or the information in this publication. Unless otherwise disclosed herein, the analysts who authored this report have not received any compensation from the subject companies in the past 12 months. If this publication contains recommendations, they are general recommendations that were prepared independently of any other interests, including those of Barclays and/or its affiliates, and/or the subject companies. This publication does not contain personal investment recommendations or investment advice or take into account the individual financial circumstances or investment objectives of the clients who receive it. Barclays is not a fiduciary to any recipient of this publication. The securities and other investments discussed herein may not be suitable for all investors and may not be available for purchase in all jurisdictions. The United States imposed sanctions on certain Chinese companies (<https://home.treasury.gov/policy-issues/financial-sanctions/sanctions-programs-and-country-information/chinese-military-companies-sanctions>), which may restrict U.S. persons from purchasing securities issued by those companies. Investors must independently evaluate the merits and risks of the investments discussed herein, including any sanctions restrictions that may apply, consult any independent advisors they believe necessary, and exercise independent judgment with regard to any investment decision. The value of and income from any investment may fluctuate from day to day as a result of changes in relevant economic markets (including changes in market liquidity). The information herein is not intended to predict actual results, which may differ substantially from those reflected. Past performance is not necessarily indicative of future results. The information provided does not constitute a financial benchmark and should not be used as a submission or contribution of input data for the purposes of determining a financial benchmark.

This publication is not investment company sales literature as defined by Section 270.24(b) of the US Investment Company Act of 1940, nor is it intended to constitute an offer, promotion or recommendation of, and should not be viewed as marketing (including, without limitation, for the purposes of the UK Alternative Investment Fund Managers Regulations 2013 (SI 2013/1773) or AIFMD (Directive 2011/61)) or pre-marketing (including, without limitation, for the purposes of Directive (EU) 2019/1160) of the securities, products or issuers that are the subject of this report.

Third Party Distribution: Any views expressed in this communication are solely those of Barclays and have not been adopted or endorsed by any third party distributor.

United Kingdom: This document is being distributed (1) only by or with the approval of an authorised person (Barclays Bank PLC) or (2) to, and is directed at (a) persons in the United Kingdom having professional experience in matters relating to investments and who fall within the definition of "investment professionals" in Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (the "Order"); or (b) high net worth companies, unincorporated associations and partnerships and trustees of high value trusts as described in Article 49(2) of the Order; or (c)

other persons to whom it may otherwise lawfully be communicated (all such persons being "Relevant Persons"). Any investment or investment activity to which this communication relates is only available to and will only be engaged in with Relevant Persons. Any other persons who receive this communication should not rely on or act upon it. Barclays Bank PLC is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority and is a member of the London Stock Exchange.

European Economic Area ("EEA"): This material is being distributed to any "Authorised User" located in a Restricted EEA Country by Barclays Bank Ireland PLC. The Restricted EEA Countries are Austria, Bulgaria, Estonia, Finland, Hungary, Iceland, Liechtenstein, Lithuania, Luxembourg, Malta, Portugal, Romania, Slovakia and Slovenia. For any other "Authorised User" located in a country of the European Economic Area, this material is being distributed by Barclays Bank PLC. Barclays Bank Ireland PLC is a bank authorised by the Central Bank of Ireland whose registered office is at 1 Molesworth Street, Dublin 2, Ireland. Barclays Bank PLC is not registered in France with the Autorité des marchés financiers or the Autorité de contrôle prudentiel. Authorised User means each individual associated with the Client who is notified by the Client to Barclays and authorised to use the Research Services. The Restricted EEA Countries will be amended if required.

Finland: Notwithstanding Finland's status as a Restricted EEA Country, Research Services may also be provided by Barclays Bank PLC where permitted by the terms of its cross-border license.

Americas: The Investment Bank of Barclays Bank PLC undertakes U.S. securities business in the name of its wholly owned subsidiary Barclays Capital Inc., a FINRA and SIPC member. Barclays Capital Inc., a U.S. registered broker/dealer, is distributing this material in the United States and, in connection therewith accepts responsibility for its contents. Any U.S. person wishing to effect a transaction in any security discussed herein should do so only by contacting a representative of Barclays Capital Inc. in the U.S. at 745 Seventh Avenue, New York, New York 10019.

Non-U.S. persons should contact and execute transactions through a Barclays Bank PLC branch or affiliate in their home jurisdiction unless local regulations permit otherwise.

This material is distributed in Canada by Barclays Capital Canada Inc., a registered investment dealer, a Dealer Member of Canadian Investment Regulatory Organization (www.ciro.ca), and a Member of the Canadian Investor Protection Fund (CIPF).

This material is distributed in Mexico by Barclays Bank Mexico, S.A. and/or Barclays Capital Casa de Bolsa, S.A. de C.V. This material is distributed in the Cayman Islands and in the Bahamas by Barclays Capital Inc., which it is not licensed or registered to conduct and does not conduct business in, from or within those jurisdictions and has not filed this material with any regulatory body in those jurisdictions.

Japan: This material is being distributed to institutional investors in Japan by Barclays Securities Japan Limited. Barclays Securities Japan Limited is a joint-stock company incorporated in Japan with registered office of 6-10-1 Roppongi, Minato-ku, Tokyo 106-6131, Japan. It is a subsidiary of Barclays Bank PLC and a registered financial instruments firm regulated by the Financial Services Agency of Japan. Registered Number: Kanto Zaimukyokuchou (kinsho) No. 143.

Asia Pacific (excluding Japan): Barclays Bank PLC, Hong Kong Branch is distributing this material in Hong Kong as an authorised institution regulated by the Hong Kong Monetary Authority. Registered Office: 41/F, Cheung Kong Center, 2 Queen's Road Central, Hong Kong.

All Indian securities-related research and other equity research produced by Barclays' Investment Bank are distributed in India by Barclays Securities (India) Private Limited (BSIPL). BSIPL is a company incorporated under the Companies Act, 1956 having CIN U67120MH2006PTC161063. BSIPL is registered and regulated by the Securities and Exchange Board of India (SEBI) as a Research Analyst: INH000001519; Portfolio Manager INP000002585; Stock Broker INZ000269539 (member of NSE and BSE); Depository Participant with the National Securities & Depositories Limited (NSDL): DP ID: IN-DP-NSDL-299-2008; Investment Adviser: INA000000391. BSIPL is also registered as a Mutual Fund Advisor having AMFI ARN No. 53308. The registered office of BSIPL is at 208, Ceejay House, Shivsagar Estate, Dr. A. Besant Road, Worli, Mumbai – 400 018, India. Telephone No: +91 22 67196399. Fax number: +91 22 67196399. Any other reports produced by Barclays' Investment Bank are distributed in India by Barclays Bank PLC, India Branch, an associate of BSIPL in India that is registered with Reserve Bank of India (RBI) as a Banking Company under the provisions of The Banking Regulation Act, 1949 (Regn No BOM43) and registered with SEBI as Merchant Banker (Regn No INM000002129) and also as Banker to the Issue (Regn No INBI00000950). Barclays Investments and Loans (India) Limited, registered with RBI as Non Banking Financial Company (Regn No RBI CoR-07-00258), and Barclays Wealth Trustees (India) Private Limited, registered with Registrar of Companies (CIN U93000MH2008PTC188438), are associates of BSIPL in India that are not authorised to distribute any reports produced by Barclays' Investment Bank.

This material is distributed in Singapore by the Singapore Branch of Barclays Bank PLC, a bank licensed in Singapore by the Monetary Authority of Singapore. For matters in connection with this material, recipients in Singapore may contact the Singapore branch of Barclays Bank PLC, whose registered address is 10 Marina Boulevard, #23-01 Marina Bay Financial Centre Tower 2, Singapore 018983.

This material, where distributed to persons in Australia, is produced or provided by Barclays Bank PLC.

This communication is directed at persons who are a "Wholesale Client" as defined by the Australian Corporations Act 2001.

Please note that the Australian Securities and Investments Commission (ASIC) has provided certain exemptions to Barclays Bank PLC (BBPLC) under paragraph 911A(2)(l) of the Corporations Act 2001 from the requirement to hold an Australian financial services licence (AFSL) in respect of financial services provided to Australian Wholesale Clients, on the basis that BBPLC is authorised by the Prudential Regulation Authority of the United Kingdom (PRA) and regulated by the Financial Conduct Authority (FCA) of the United Kingdom and the PRA under United Kingdom laws. The United Kingdom has laws which differ from Australian laws. To the extent that this communication involves the provision of financial services by BBPLC to Australian Wholesale Clients, BBPLC relies on the relevant exemption from the requirement to hold an AFSL. Accordingly, BBPLC does not hold an AFSL.

This communication may be distributed to you by either: (i) Barclays Bank PLC directly or (ii) Barrenjoey Markets Pty Limited (ACN 636 976 059, "Barrenjoey"), the holder of Australian Financial Services Licence (AFSL) 521800, a non-affiliated third party distributor, where clearly identified to you by Barrenjoey. Barrenjoey is not an agent of Barclays Bank PLC.

This material, where distributed in New Zealand, is produced or provided by Barclays Bank PLC. Barclays Bank PLC is not registered, filed with or approved by any New Zealand regulatory authority. This material is not provided under or in accordance with the Financial Markets Conduct Act of 2013 ("FMCA"), and is not a disclosure document or "financial advice" under the FMCA. This material is distributed to you by either: (i) Barclays Bank PLC directly or (ii) Barrenjoey Markets Pty Limited ("Barrenjoey"), a non-affiliated third party distributor, where clearly identified to you by Barrenjoey. Barrenjoey is not an agent of Barclays Bank PLC. This material may only be distributed to "wholesale investors" that meet the "investment business", "investment activity", "large", or "government agency" criteria specified in Schedule 1 of the FMCA.

Middle East: Nothing herein should be considered investment advice as defined in the Israeli Regulation of Investment Advisory, Investment Marketing and Portfolio Management Law, 1995 ("Advisory Law"). This document is being made to eligible clients (as defined under the Advisory Law) only. Barclays Israeli branch previously held an investment marketing license with the Israel Securities Authority but it cancelled such license on 30/11/2014 as it solely provides its services to eligible clients pursuant to available exemptions under the Advisory Law, therefore a license with the Israel Securities Authority is not required. Accordingly, Barclays does not maintain an insurance coverage pursuant to the Advisory Law.

This material is distributed in the United Arab Emirates (including the Dubai International Financial Centre) and Qatar by Barclays Bank PLC. Barclays Bank PLC in the Dubai International Financial Centre (Registered No. 0060) is regulated by the Dubai Financial Services Authority (DFSA). Principal place of business in the Dubai International Financial Centre: The Gate Village, Building 4, Level 4, PO Box 506504, Dubai, United Arab Emirates. Barclays Bank PLC-DIFC Branch, may only undertake the financial services activities that fall within the scope of its existing DFSA licence. Related financial products or services are only available to Professional Clients, as defined by the Dubai Financial Services Authority. Barclays Bank PLC in the UAE is regulated by the Central Bank of the UAE and is licensed to conduct business activities as a branch of a commercial bank incorporated outside the UAE in Dubai (Licence No.: 13/1844/2008, Registered Office: Building No. 6, Burj Dubai Business Hub, Sheikh Zayed Road, Dubai City) and Abu Dhabi (Licence No.: 13/952/2008, Registered Office: Al Jazira Towers, Hamdan Street, PO Box 2734, Abu Dhabi). This material does not constitute or form part of any offer to issue or sell, or any solicitation of any offer to subscribe for or purchase, any securities or investment products in the UAE (including the Dubai International Financial Centre) and accordingly should not be construed as such. Furthermore, this information is being made available on the basis that the recipient acknowledges and understands that the entities and securities to which it may relate have not been approved, licensed by or registered with the UAE Central Bank, the Dubai Financial Services Authority or any other relevant licensing authority or governmental agency in the UAE. The content of this report has not been approved by or filed with the UAE Central Bank or Dubai Financial Services Authority. Barclays Bank PLC in the Qatar Financial Centre (Registered No. 00018) is authorised by the Qatar Financial Centre Regulatory Authority (QFCRA). Barclays Bank PLC-QFC Branch may only undertake the regulated activities that fall within the scope of its existing QFCRA licence. Principal place of business in Qatar: Qatar Financial Centre, Office 1002, 10th Floor, QFC Tower, Diplomatic Area, West Bay, PO Box 15891, Doha, Qatar. Related financial products or services are only available to Business Customers as defined by the Qatar Financial Centre Regulatory Authority.

Russia: This material is not intended for investors who are not Qualified Investors according to the laws of the Russian Federation as it might contain information about or description of the features of financial instruments not admitted for public offering and/or circulation in the Russian Federation and thus not eligible for non-Qualified Investors. If you are not a Qualified Investor according to the laws of the Russian Federation, please dispose of any copy of this material in your possession.

Environmental, Social, and Governance ('ESG') Related Research: There is currently no globally accepted framework or definition (legal, regulatory or otherwise) of, nor market consensus as to what constitutes, an 'ESG', 'green', 'sustainable', 'climate-friendly' or an equivalent company, investment, strategy or consideration or what precise attributes are required to be eligible to be categorised by such terms. This means there are different ways to evaluate a company or an investment and so different values may be placed on certain ESG credentials as well as adverse ESG-related impacts of companies and ESG controversies. The evolving nature of ESG considerations, models and methodologies means it can be challenging to definitively and universally classify a company or investment under an ESG label and there may be areas where such companies and investments could improve or where adverse ESG-related impacts or ESG controversies exist. The evolving nature of sustainable finance related regulations and the development of jurisdiction-specific regulatory criteria also means that there is likely to be a degree of divergence as to the interpretation of such terms in the market. We expect industry guidance, market practice, and regulations in this field to continue to evolve. Any references to 'sustainable', 'sustainability', 'green', 'social', 'ESG', 'ESG considerations', 'ESG factors', 'ESG issues' or other similar or related terms in this document are as used in our public disclosures and not to any jurisdiction-specific regulatory definition or other interpretation of these terms unless specified otherwise.

IRS Circular 230 Prepared Materials Disclaimer: Barclays does not provide tax advice and nothing contained herein should be construed to be tax advice. Please be advised that any discussion of U.S. tax matters contained herein (including any attachments) (i) is not intended or written to be used, and cannot be used, by you for the purpose of avoiding U.S. tax-related penalties; and (ii) was written to support the promotion or marketing of the transactions or other matters addressed herein. Accordingly, you should seek advice based on your particular circumstances from an independent tax advisor.

© Copyright Barclays Bank PLC (2024). All rights reserved. No part of this publication may be reproduced or redistributed in any manner without the prior written permission of Barclays. Barclays Bank PLC is registered in England No. 1026167. Registered office 1 Churchill Place, London, E14 5HP. Additional information regarding this publication will be furnished upon request.

U.S. Semiconductors & Semiconductor Capital Equipment

Kyle Bleustein

+1 212 526 7618

kyle.bleustein@barclays.com

BCI, US